

Research document

Basketball Royale

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Revision history

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0.2	07-02-2023	Sander Theeuwen	Added “Why?” to all questions. Added approach to all questions. Added result to first question.	Unpublished
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Distribution

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0.4	12-06-2023	Fontys
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Introduction

Making a game draw attention and keeping the players interested is the main goal of a game. Especially on Twitch where broadcasters need to keep viewers entertained, it is important that this is done in the most effective way. This document contains research to assist in the development of the game to achieve this goal.

Problem statement

Broadcasters on Twitch sometimes need a break for various reasons. Making a phone call, answering the door, going to the bathroom for example. In the meantime, viewers must be entertained without the streamer being present.

Broadcasters also hold giveaways from time to time. While they can just choose someone or come up with a way of selecting a viewer at random, a game with a winner can be effective as well.

The above two scenarios are the main reasons for broadcasters to use the game. In both cases, the viewers need to be entertained. The most effective ways to do so are explored in this research document.

How can a round based royale game be enjoyable for varying amounts of players on Twitch?

How do other games handle varying amounts of players on Twitch?

Why?

As this is most likely not the first project with this problem, knowing how others have tackled the problem can give useful insights and ideas to use for Basketball Royale. It makes brainstorming solutions easier.

Approach

Community research will show the games that are most played with large communities.

For a smaller number of players, brainstorming about different mechanics / features will be the main source of ideas.

Result

Community research

The sources were scarce, but Marbles on Stream and Words on Stream seemed to be the main competitors. Especially Marbles on Stream supports a lot of players.

In Marbles on Stream, players have a marble with their name on it. Once the game starts, all marbles are released on a track, where the first marble to get to the finish wins. There is a lot of viewer recognition, as the marbles are visible for the full duration of the game. When events happen, like marbles falling off the level or getting stuck, it gives the streamer an easy opportunity to commentate, adding even more viewer recognition.

In Words on Stream, players are given letters with which they must make words. When a player guesses a word, it is displayed in the game with the player's name above it. After a round, the scores are displayed for that round, plus the total scores. Along with viewer recognition, competition as well as teamwork plays a big role.

Brainstorming

The manager, company mentor, and I had several meetings in which we brainstormed about viewer recognition and mechanics that involved players' input. As the project can use an extension, players can click on the screen (if their browser supports it) and the game can use that input. We came up with the idea where the player could specify coordinates to throw to. To make it more exciting and arcade-like, it would be in the form of having a throwing origin point as the horizontal factor and a throw strength as the vertical factor.

What is enjoyable for a player as Twitch viewer?

Why?

The game is meant to last a few minutes while the streamer does something else like making a phone call or answering the door, or to determine a winner for a giveaway. The game must be interesting enough to invite people to participate and to keep them busy.

Game feel is extremely important. Even little adjustments like particles or screen shake can make a huge difference.

Approach

During the development, playtests are held frequently to check if the added features are fun enough to keep the players interested. Observations are particularly useful to test if the latest changes are intuitive.

Later in the development, experts from Business Goose are asked to give their opinion about the game feel and what can be done to improve it.

Result

Playtests

At the playtests, the feedback I got was always positive. Most of the time, there were some suggestions, which were implemented before the next playtest to ensure the quality is always increasing.

Expert interview

In the meetings with Business Goose, the impact of certain game events was the main topic of conversation. A demo was given of the project, with the question to look for improvements regarding game feel. Effects like particles, and screen shake were mentioned to be effective, with more effects for rarer events. For example, hitting the basket's ring could produce a tiny bit of screen shake, whereas scoring has relatively much screen shake, since it is the biggest event in the game. To add to the game feel, a typewriter effect was suggested, as it not only looks more realistic for an arcade setting, but also draws attention to that area of the screen.

What is enjoyable for a spectator as Twitch broadcaster?

Why?

As the game can also be used for determining a winner for a giveaway or a quick game in between sections of a stream, the broadcaster also needs to enjoy the game. There must be enough action for them to comment on.

Approach

Brainstorming about ways to make the game include action and other things to comment on will be the main source to answer this question. If multiple ideas are thought of, they might all be included in the product. In meetings, we deployed the game and role played to fully experience the live stream as it would in other streams.

Results

In the design phase, I brainstormed, sometimes together with the manager and company mentor, about events to make the game interesting and contain action. Water filling up the arcade hall affecting the required force to throw basketballs or fog making it hard to see could be ways to immediately get some action. The focus of the project, however, was to create a minimal viable product as quickly as possible and then pay attention to the details. Once the product was ready for its first release, a meeting was held where the idea arose to randomly throw a basketball at a window, making a crack appear. Events like these can give a broadcaster something to easily comment on.

To which extend can enjoyable features / aspects be implemented for a varying number of Twitch viewers?

Why?

With many players playing the game, it can cause performance issues if many things are happening, especially with the basketballs of the players. Some ideas gathered in the previous research questions might not be suitable for the game as they affect the performance too much.

Approach

Expert interviews are conducted to determine when the performance needs to be tested. When the performance is affected too much, the experts can guide in reducing the resources used or to give the best advice on what to do with the feature should it have too much impact.

Results

In a meeting with the company mentor, he stated that benchmarking is not necessary if it is not visible. By worrying about the performance and writing the code as efficiently as possible, more time is consumed, where the gained performance might not be worth it. I should focus on implementing features instead of always taking the resource usage into account.

Conclusion

In all cases, whether there are many players or just a few, viewer recognition stands above all. Adding features where the players can interact with the game is also a good way to keep viewers interested. Finally, random events that stand out in comparison to the regular game flow, like basketballs hitting windows, add action to the game that gets the attention of the viewers, and the broadcaster can easily comment on.

It is important that a minimal viable product is created as soon as possible, after which more features that do not affect the core loop of the game can be added. Performance wise, it is best to investigate performance issues when the issues arise instead of preemptively.